Flight Delay Prediction Using Logistic Regression and Gradient Boosting

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***Abstract*—Flight delays affect passengers, airlines, and airport operations. This project develops a machine learning pipeline to predict flight arrival delays using historical flight data. The study uses the 2013 New York City flights dataset and defines a delayed flight as one with an arrival delay of at least 15 minutes. Logistic Regression is used as a baseline model, while XGBoost is used as the gradient boosting model. The workflow includes exploratory data analysis, feature engineering, chronological data splitting, model evaluation, and performance comparison. The results show that XGBoost achieves a higher precision-recall area under the curve and recall than the baseline at the evaluated thresholds, although both models have limited discrimination performance.**

***Index Terms*—Flight delay prediction, machine learning, Logistic Regression, XGBoost, feature engineering, classification**

I. Introduction

Flight delays create inconvenience for passengers and operational challenges for airlines. Predicting delays in advance can help stakeholders understand the likelihood of disruption and support better operational planning.

Machine learning methods can identify patterns in historical flight data and use them to estimate the probability of a future delay. This project implements a classification pipeline using historical flight records, engineered temporal features, and two machine learning models.

The current work focuses on the first phase of the project: exploratory analysis, feature engineering, model training, and evaluation.

II. Problem Formulation

The task is formulated as a binary classification problem. Let A_i represent the arrival delay, in minutes, for flight i. The target variable is defined as

$$y_i = \begin{cases} 1, & A_i \geq 15, \\ 0, & A_i < 15. \end{cases} \tag{1}$$

Records with missing arrival-delay labels are removed before constructing the target. The model uses information about the scheduled flight and route to predict whether the flight will be delayed.

Actual arrival delay and other post-departure information are excluded from the input features to reduce target leakage.

III. Dataset and Preprocessing

The project uses the 2013 New York City flights dataset, which contains flight records associated with New York-area airports.

The raw dataset contains 336,776 rows. After removing records with missing arrival-delay labels, 327,346 rows remain for analysis. The resulting processed dataset contains 18 columns: one scheduled datetime column, 16 engineered input features, and the binary target.

The observed positive-class rate is approximately 24.47%.

Preprocessing includes removing records without target labels, constructing the binary delay target, converting scheduled time information into datetime values, and arranging observations chronologically. A chronological split is used, with 70% of the data for training, 15% for validation, and the final 15% for testing.

IV. Feature Engineering

Features are designed to represent scheduled timing and flight characteristics. They include month, day, hour, minute, day of week, day of year, and flight distance. The carrier, origin airport, and destination airport are also included.

Cyclical time features represent periodic relationships. For example, the hour of day is encoded as

$$h_{\sin} = \sin\left(\frac{2\pi h}{24}\right), \tag{2}$$

$$h_{\cos} = \cos\left(\frac{2\pi h}{24}\right). \tag{3}$$

Similar sine and cosine transformations are used for day of week and day of year. These representations allow values near the end and beginning of a cycle to remain close in feature space.

V. Methodology

A. Logistic Regression

Logistic Regression is used as the baseline classifier. It estimates the probability of the positive class using a logistic function:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp(-(\mathbf{w}^T\mathbf{x} + b))}. \tag{4}$$

A classification threshold converts the estimated probability into a binary prediction.

B. XGBoost

XGBoost is a gradient boosting method that combines multiple decision trees. Each successive tree contributes to improving the model's objective. It can capture nonlinear relationships and interactions between flight timing, distance, carrier, and airport features.

The project uses 400 estimators, a maximum tree depth of 6, a learning rate of 0.05, subsampling of 0.8, and column subsampling of 0.8. The binary logistic objective is used for classification.

C. Evaluation Metrics

Model performance is assessed using precision, recall, F1-score, accuracy, area under the precision-recall curve (PR-AUC), area under the receiver operating characteristic curve (ROC-AUC), and the Brier score.

Precision measures the fraction of predicted delays that are actual delays. Recall measures the fraction of actual delays identified by the model. The F1-score is the harmonic mean of precision and recall. PR-AUC and ROC-AUC assess ranking performance, while the Brier score measures the mean squared error of predicted probabilities.

VI. Experimental Results

Table **??** presents the saved-model evaluation results on the final 15% of the processed dataset. The reported metrics use the thresholds stored with the respective model bundles.

XGBoost obtains a PR-AUC of 0.3338 and ROC-AUC of 0.5985, compared with 0.3084 and 0.5682 for Logistic Regression. At its evaluated threshold of 0.3000, XGBoost achieves a recall of 0.5054 and an F1-score of 0.4040.

Logistic Regression produces no positive predictions at its saved threshold of approximately 0.7223, resulting in zero precision, recall, and F1-score under the evaluation convention used. Its accuracy is 0.7331, but accuracy alone is not sufficient to assess a model on an imbalanced classification problem.

The Brier scores are similar for the two models. These results suggest that XGBoost performs better on the reported ranking metrics and identifies more delayed flights at the evaluated threshold. However, the overall discrimination performance remains limited.

VII. Discussion

The results demonstrate the importance of selecting appropriate evaluation metrics and classification thresholds. A high accuracy score does not necessarily indicate effective delay detection when the target classes are imbalanced.

XGBoost provides a more useful balance between precision and recall at the threshold used in this evaluation. Nevertheless, the results indicate that additional work is required before the system can be used for operational decision-making.

Threshold selection should reflect the intended application. For example, a system designed to identify as many delayed flights as possible may prioritize recall, while a system intended to minimize false alarms may prioritize precision.

VIII. Limitations and Future Work

This study has several limitations. The dataset represents flights from a specific region and year, so the findings may not generalize to other airports, airlines, or periods. The models also have limited ranking performance, indicating that the current features and approaches do not fully explain flight delays.

Future work may include improved threshold selection, further hyperparameter tuning, richer operational and weather information, and evaluation across different time periods. Later project phases may investigate temporal deep learning methods and offline rescheduling recommendations. These extensions are outside the scope of the current Phase 1 implementation.

IX. Conclusion

This project implements a flight-delay classification pipeline using historical flight records, temporal feature engineering, Logistic Regression, and XGBoost. The processed dataset contains 327,346 labeled flights, with approximately 24.47% classified as delayed.

In the reported evaluation, XGBoost outperforms Logistic Regression on PR-AUC and ROC-AUC and achieves a recall of approximately 0.505 at its evaluated threshold. The findings also show why accuracy should be considered alongside precision, recall, F1-score, and probability-based metrics. Further improvements are needed to develop a more reliable flight-delay prediction system.

References

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TABLE I
COMPARISON OF MODEL PERFORMANCE

Model	Threshold	PR-AUC	ROC-AUC	Precision	Recall	F1-score	Accuracy	Brier Score
Logistic Regression	0.7223	0.3084	0.5682	0.0000	0.0000	0.0000	0.7331	0.2092
XGBoost	0.3000	0.3338	0.5985	0.3364	0.5054	0.4040	0.6019	0.2091